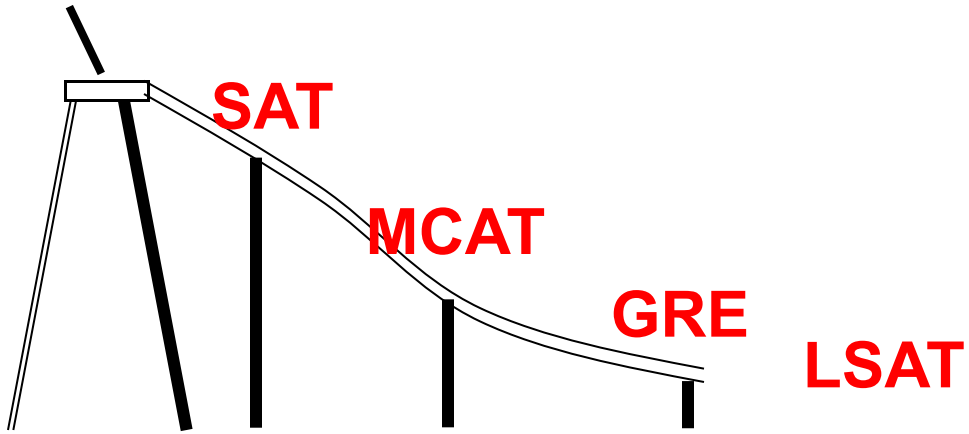
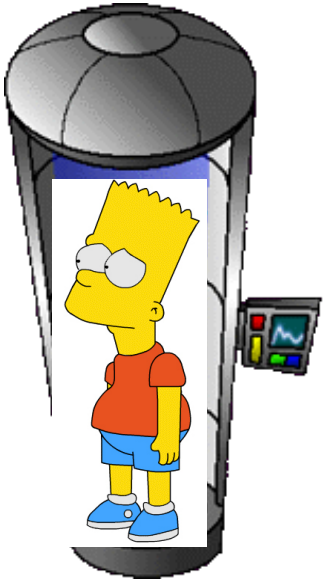
Test Slide



Quantum Communication





Dave Bacon

Department of Physics

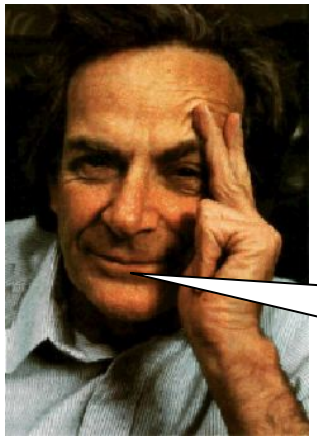
Institute for Quantum Information

Caltech





Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Anyone who is not shocked by quantum theory has not understood it.



Niels Bohr
Nobel Prize 1922



They understand it.
They use it.
But they don’t really believe it?

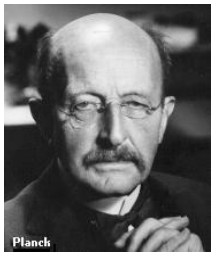
Quantum History

Inventin

Worrying

Acceptin g

1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000



Max Planck


Niels Bohr

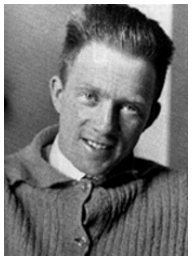

Erwin Schrodinger

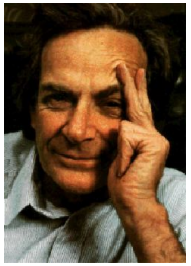

John Von Neumann


John Bell

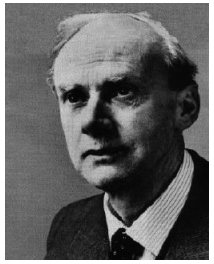

Peter Shor


Louis de Broglie


Werner Heisenberg


Richard Feynman


John Archibald Wheeler


P.A.M. Dirac

Acceptance

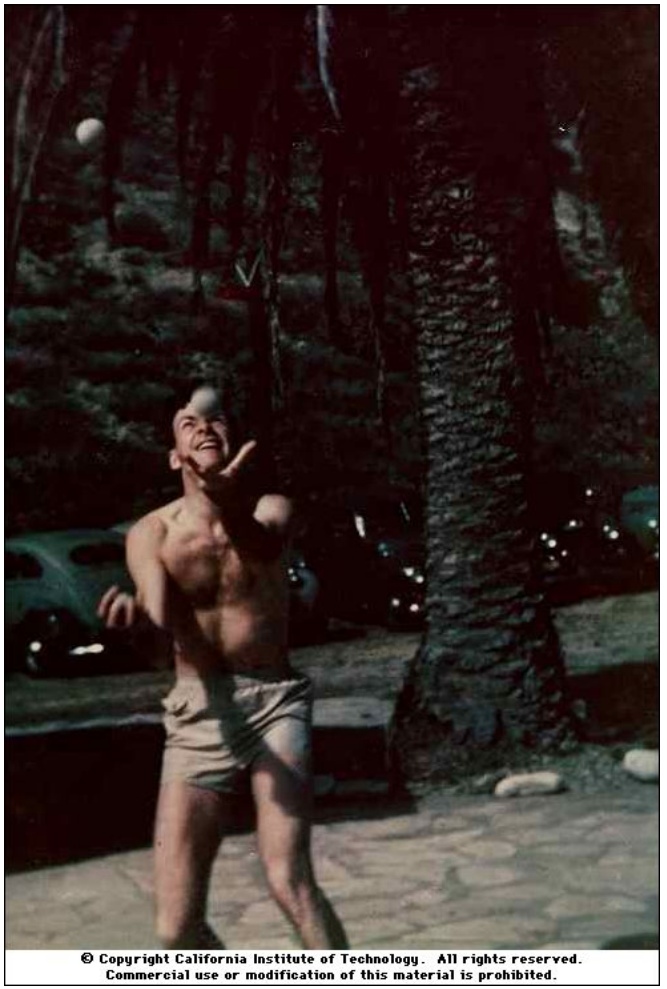
on quantum theory:

"Do not ask yourself, if you can possibly avoid that, 'how can it be like that?' because you will lead yourself down a blind alley in which no one has ever escaped."

"For a successful technology, nature must take precedence over public relations, for nature cannot be fooled."

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer?

It's not a Turing machine but a machine of a different kind" (1981)

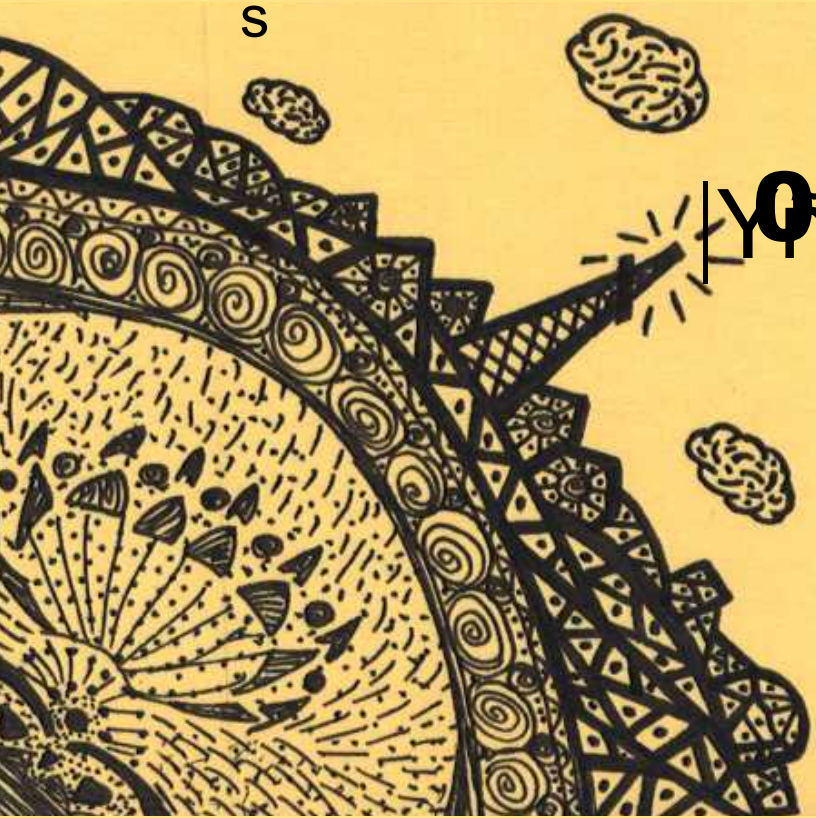

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Richard Feynman

Quantum Radio

probability	
0	p_0
1	$1 - p_0$

bit
s



amplitude
s

$$|\psi\rangle \triangleq \alpha|0\rangle + \beta|1\rangle$$

qubits

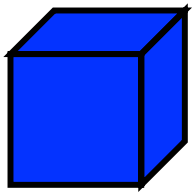




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From Here to There







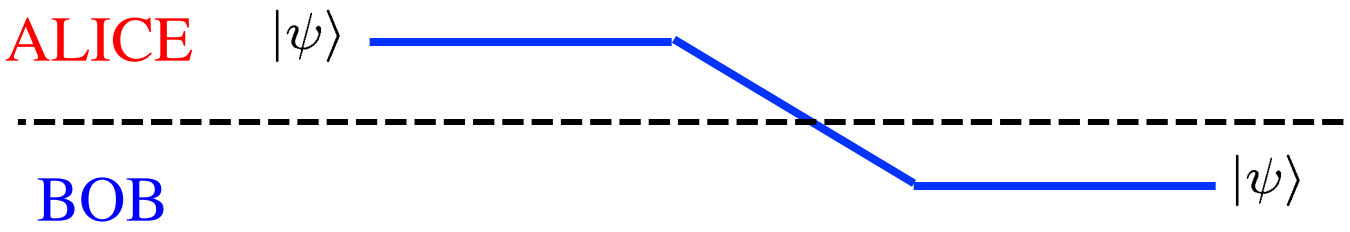
ALICE

some vast distance

(The shortest distance between two points is under construction.)

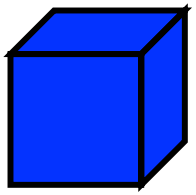
BOB

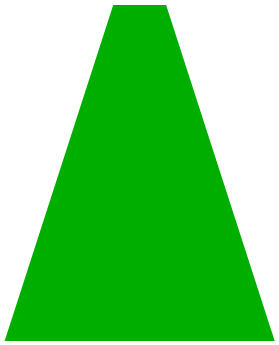
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



From Here to There









ALICE

some vast distance

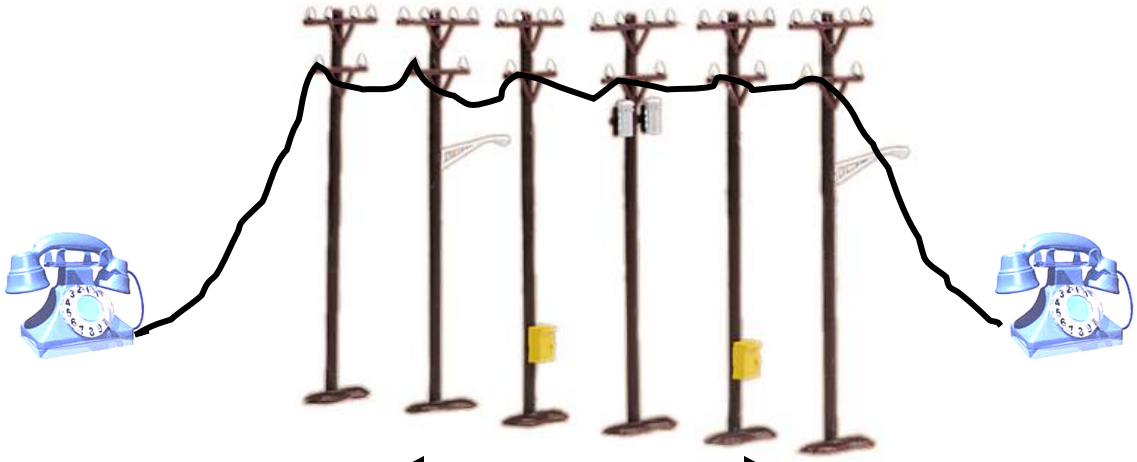
(The shortest distance between two points is under construction.)

BOB

Question: Other methods to transport quantum information?

Silly Question, Silly Answer







ALICE

BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits



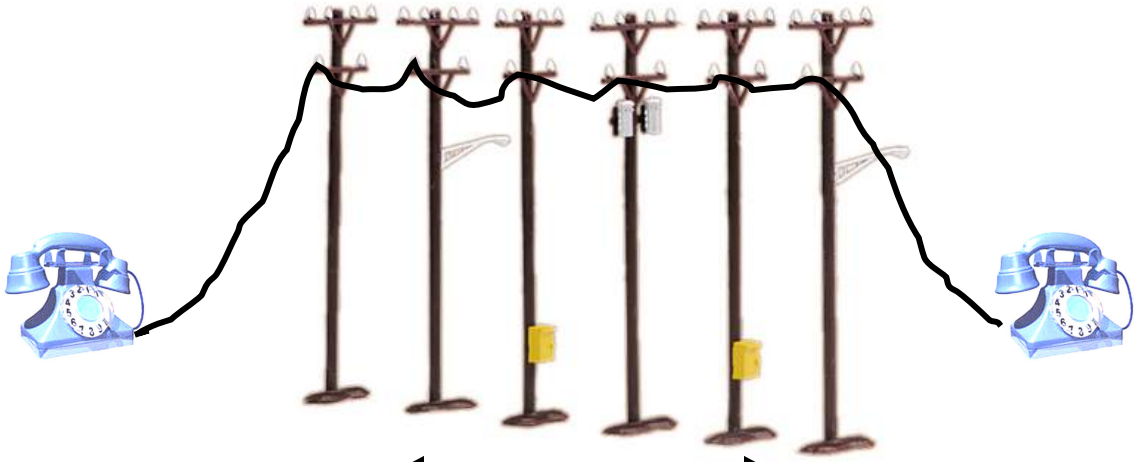
two problems:

infinite number of bits to specify $|\psi\rangle$

cannot perfectly learn unknown $|\psi\rangle$

Silly Question?







ALICE

BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits

+

shared entanglement

A Silly Question?

Entanglement



ALICE

$|\psi_A\rangle = \alpha_A|0\rangle + \beta_B|1\rangle$ $|\psi_B\rangle = \alpha_B|0\rangle + \beta_B|1\rangle$



BOB

$|\psi_A\rangle \otimes |\psi_B\rangle$

separable

$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$

entangled

Entanglement Is Like…

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$



ALICE

measure in $|0\rangle$, $|1\rangle$ basis

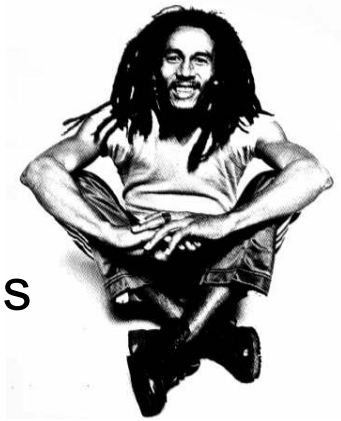
outcome

50% $|0\rangle \otimes |0\rangle$

50% $|1\rangle \otimes |1\rangle$

random correlated bits

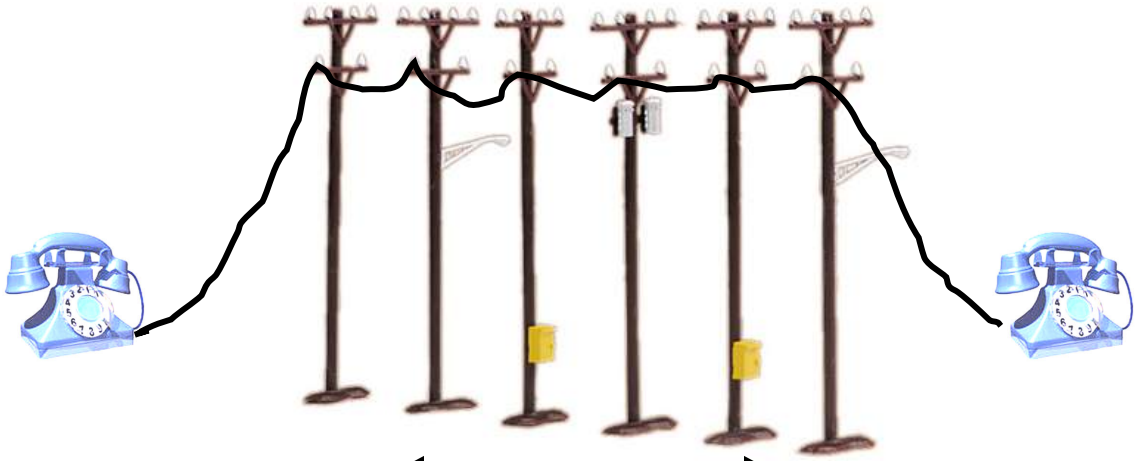
measure in $|0\rangle$, $|1\rangle$ basis



BOB

Silly Question?







ALICE

BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits

+

random correlated bits

classical bits

random correlated bits

A Silly Question!

Entanglement Is Like…

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$



ALICE

measure in $|0\rangle$, $|1\rangle$ basis

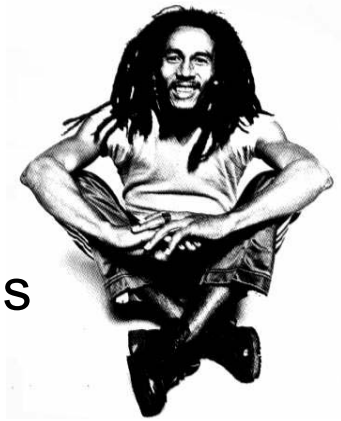
outcome

50% $|0\rangle \otimes |0\rangle$

50% $|1\rangle \otimes |1\rangle$

random correlated bits

measure in $|0\rangle$, $|1\rangle$ basis



BOB

Is It Like Random Correlated Bits?

Bell's Question

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

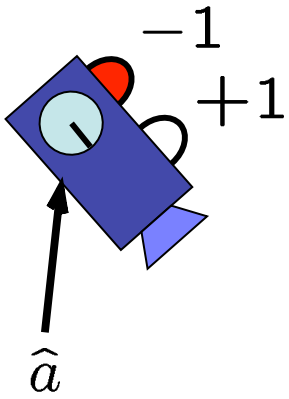


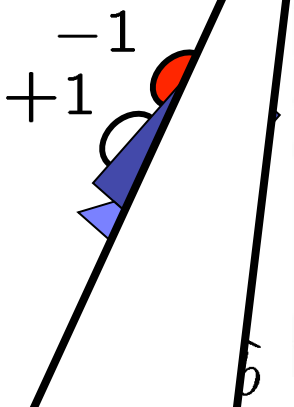
John Bell

A

B









ALICE

$$\langle A(\hat{a})B(\hat{b})\rangle = -\hat{a}\cdot\hat{b}$$

BOB

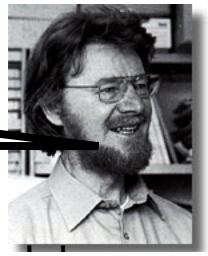
If $\hat{a} = \hat{b}$ then $\langle AB\rangle = -1$, i.e. outputs are "anti"-correlated

$A =$ -1

$A =$ $+1$

Can these correlations be explained by random correlated bits?

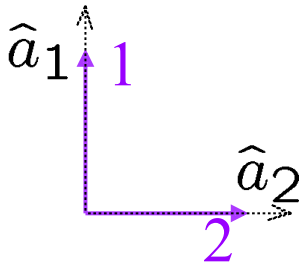
Bell Thinks



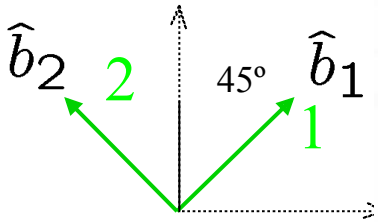
$$\langle A(\hat{a})B(\hat{b})\rangle = -\hat{a}\cdot\hat{b}$$



ALICE



	1	2
1	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$



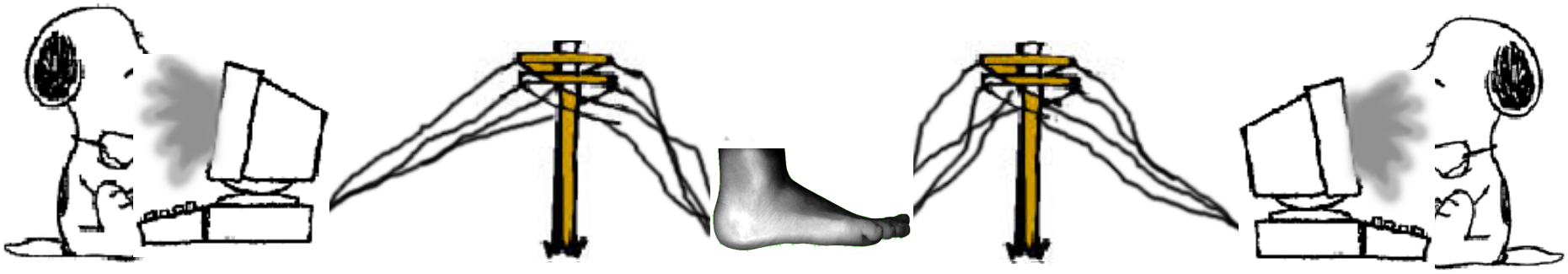


BOB

$$\langle A(\hat{a}_1)B(\hat{b}_1)\rangle + \langle A(\hat{a}_2)B(\hat{b}_1)\rangle + \langle A(\hat{a}_1)B(\hat{b}_2)\rangle - \langle A(\hat{a}_2)B(\hat{b}_2)\rangle$$

Quantum: $-\frac{4}{\sqrt{2}} = -2\sqrt{2}$

Communication Breakdown



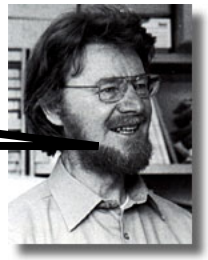
1. Initial unlimited classical communication between computers.
2. Communication lines cuts. **random correlated bits**
3. Input into each computer.
4. Output by each computer.

Repeat

Build up table of input-output correlations

What sort of correlations can be produced?

Bell's Thinks II



"Convex"–iftying a probabilistic protocol

random variables could all be generated in first step of protocol

probabilistic protocol = deterministic protocol with certain probabilities

$$A(\hat{a}_i) \in \{+1, -1\} \qquad B(\hat{b}_i) \in \{+1, -1\}$$

Alice's output and measurement choice
cannot depend on
Bob's output and measurement choice (and vice versa)

$$\langle A(\hat{a}_1)B(\hat{b}_1)\rangle + \langle A(\hat{a}_2)B(\hat{b}_1)\rangle + \langle A(\hat{a}_1)B(\hat{b}_2)\rangle - \langle A(\hat{a}_2)B(\hat{b}_2)\rangle$$

deterministic protocol

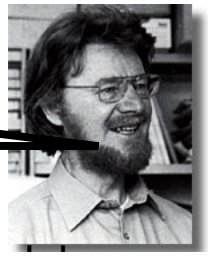
$$A(\hat{a}_1)B(\hat{b}_1) + A(\hat{a}_2)B(\hat{b}_1) + A(\hat{a}_1)B(\hat{b}_2) - A(\hat{a}_2)B(\hat{b}_2)$$

$$A(\hat{a}_1)[B(\hat{b}_1) + B(\hat{b}_2)] + A(\hat{a}_2)[B(\hat{b}_1) - B(\hat{b}_2)] = \pm 2$$

average over shared random bits

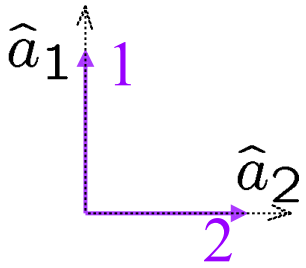
$$|A(\hat{a}_1)[B(\hat{b}_1) + B(\hat{b}_2)] + A(\hat{a}_2)[B(\hat{b}_1) - B(\hat{b}_2)]| \leq 2$$

Bell Answers

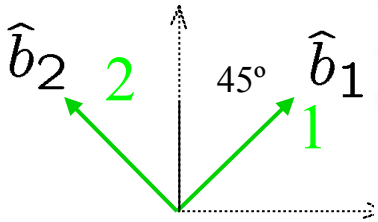


$$\langle A(\hat{a})B(\hat{b})\rangle = -\hat{a}\cdot\hat{b}$$





	1	2
1	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$





ALICE

BOB

$$\langle A(\hat{a}_1)B(\hat{b}_1)\rangle + \langle A(\hat{a}_2)B(\hat{b}_1)\rangle + \langle A(\hat{a}_1)B(\hat{b}_2)\rangle - \langle A(\hat{a}_2)B(\hat{b}_2)\rangle$$

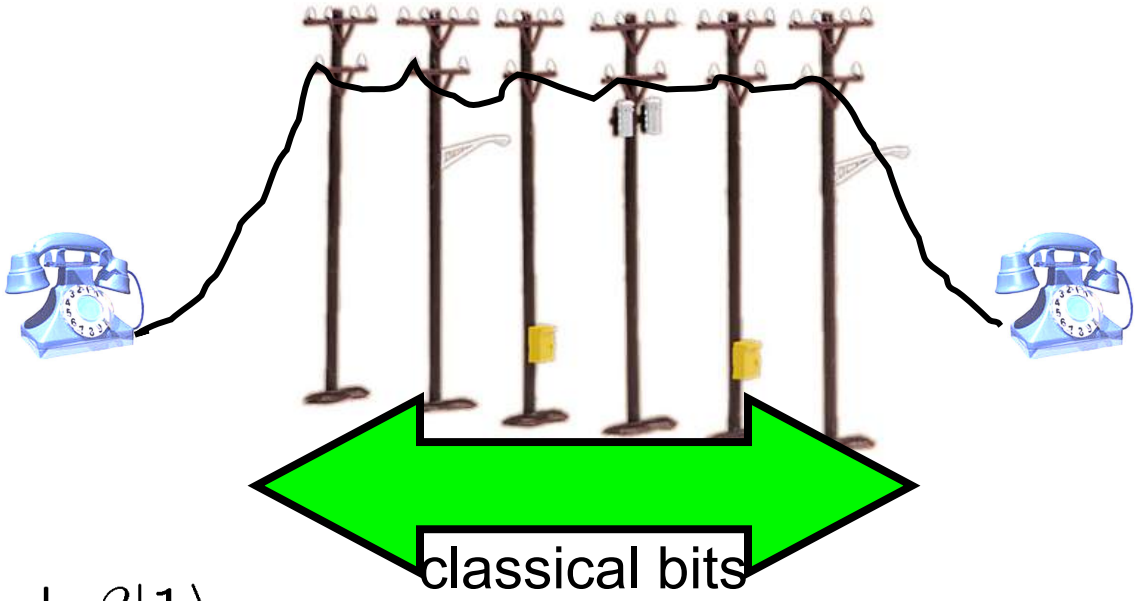
Quantum: $-\frac{4}{\sqrt{2}} = -2\sqrt{2}$

With correlated random bits: between -2 and $+2$

Quantum correlations are DIFFERENT from correlated random bits!

From Here to There







ALICE

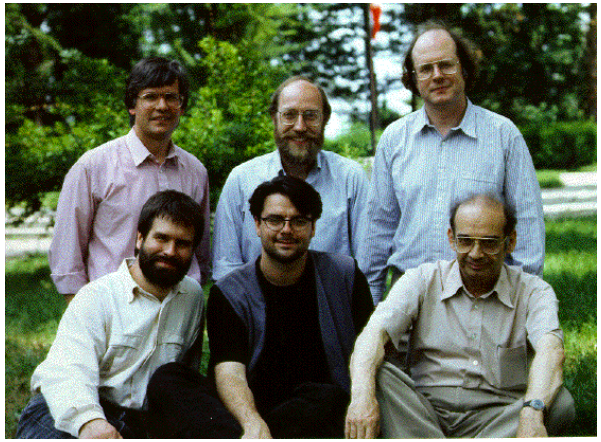
BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits

+

shared entanglement



R. Jozsa, W. Wootters, C.H. Bennet
G. Brassard, C. Crepeau, and A. Peres